

George Zhang

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SKILLS

- **Languages:** Python, Java, C/C++, Kotlin, HTML, SQL
- **Frameworks & Libraries:** OpenCV, MediaPipe, Numpy, Android SDK, Room Persistence Library
- **Tools & Platforms:** Git, Linux, Google Play Console, Android Studio
- **Concepts:** Algorithms & Data Structures, Object-Oriented Design, Computer Vision, Full-Stack Development

EDUCATION

University of Waterloo
Bachelor of Computer Science

Sep 2024 – May 2029
Waterloo, ON

WORK EXPERIENCE

Undergraduate Research Assistant
University of British Columbia

Jun 2025 – Aug 2025
Vancouver, BC

- Took charge of the development of two full-stack applications including HandEase and Palm & Plant to assist in Parkinson's rehabilitation, projecting to aid medically patients in hospitals across 10+ countries.
- Developed a computer vision pipeline using Python, OpenCV, and MediaPipe – AI hand library to detect and collect rehabilitation data on 7 hand exercise gestures of patients, resulting in ~90% accuracy for medical analysis.
- Optimized Apps to run in real time (<100ms latency) on local hardware, delivering immediate results and feedback to create polished and consistent user experiences.
- Designed dashboards with activity heatmap and daily metrics using Python to improve monitoring efficiency for practitioners by 30% and streamline long-term medical care for patients.
- Led end-to-end project lifecycle from requirements, development, testing, and deployment, which was presented in the 11th Singapore International Parkinson Disease & Movement Disorders Symposium.
- Engineered 10 applications with 4 team members and updated weekly progress to clinicians at the Djavad Mowafaghian Centre for Brain Health, receiving positive reviews.

Math & Reading Instructor
Kumon Inc.

Aug 2021 – May 2024
Vancouver, BC

- Invited by Kumon Supervisor to join their team due to outstanding performance in Math and Literature
- Mentored 100+ students in math/reading, marked their homework, adapted lessons to individual needs, and provided optimal and precise guidance
- Managed class flow and delivered feedback according to students personal understanding and performance, leading to students' significant improvement in tests and classroom behavior

PROJECTS

My Fridge App | Kotlin | Android SDK | Room Database

Jun 2023 – Aug 2023

- Designed an Android application managing fridge inventory and published on Google Play, reducing food waste by 20% and inducing healthy eating habits.
- Implemented Room DB + push notifications reminding users of expiry dates and track shopping lists with >95% reliability, constructing efficient food usage and enhancing shopping experiences.

RoboCup Junior Maze Rescue - Captain | Python | OpenMV | LiDAR | Algorithms

Jun 2022 – Jul 2023

- Architected a prioritized maze navigation algorithm with modified Dijkstra's and LiDAR-based positioning for multi-level point-based maps.
- Facilitated CV logic for letter and color detection with >90% accuracy, reacting to maze events effectively.
- Designed a robot achieving 85% maze success rate and completing runs <2 mins, which led team to place 2nd in the Supergroup Category and 17th in the Individual Component out of 50+ teams at RoboCup International 2023.

Game Development Portfolio | Java

Feb 2021 – Jul 2022

- Created 3 Java games of ~3000 LOC to strengthen OOP/design skills, and implemented DFS group capture for Go, randomized cascade reveal for Minesweeper, and custom 2D physics in platformer parkour game.

AWARDS AND ACHIEVEMENTS

- **International Collegiate Programming Contest Local Competition (Waterloo)** | Rank 28 out of ~100 Oct 2024
- **RoboCup International Junior Maze Rescue** | 2nd Place Supergroup, 17th Individual Jul 2023
- **Avogadro's Chemistry Contest** | Certificate of Distinction May 2023
- **AIME (American Invitational Mathematics Examination) Qualifier** (top >10% of 150,000) Nov 2023
- **COMC (Canadian Open Mathematics Challenge) Distinction** May 2023